



TRAINING PROGRAMME

Kubernetes Administration (CKA)

Install, secure, upgrade and troubleshoot Kubernetes clusters

36 HOURS

FOR Engineers who run Kubernetes clusters and are preparing the CKA exam

PREPARES FOR Certified Kubernetes Administrator (CKA, 2025 curriculum)



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Overview

This programme prepares participants to administer Kubernetes clusters and to sit the Certified Kubernetes Administrator exam on its 2025 curriculum. They install a highly available cluster with kubeadm, upgrade it and back up etcd, secure it with RBAC and security contexts, control scheduling, configure cluster networking from the CNI to the Gateway API, manage storage, and extend the cluster with Helm, Kustomize, custom resources and operators. Every lab ends with a troubleshooting step, a timed mock exam measures readiness, and the programme closes with a final project: a cluster built, secured, upgraded and repaired by each team. etcd backup and restore is taught as a production skill; it is no longer a CKA objective.

At a glance

#	PHASE	FOCUS	HOURS
01	Cluster Installation & Lifecycle	kubeadm, highly available control plane, upgrades, etcd backup	6 h
02	Security & Scheduling	RBAC, ServiceAccounts, security contexts, scheduling controls	6 h
03	Networking, Storage & Extensions	CNI, Services and DNS, NetworkPolicies, Gateway API, storage, Helm, CRDs	6 h
04	Troubleshooting & Exam Preparation	Node, pod and network troubleshooting, timed mock exam	6 h
05	Final Project	A highly available cluster, built, secured, upgraded and repaired	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
	Total		36 h

Key facts

AUDIENCE	Engineers who run Kubernetes clusters and are preparing the CKA exam
PREREQUISITES	Kubernetes user level (course: Docker & Kubernetes): Pods, Deployments, Services, Helm. Using Kubernetes is not taught again; this course is about running it
DURATION	36 h: 30 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs at the command line; one timed mock exam
LAB ENVIRONMENT	Runs on premises: one lab server per team of two (64 GB RAM), provided by the organiser, hosting the Ubuntu virtual machines of each cluster (three control-plane nodes, two workers, a load balancer); open-source software only; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase, a timed mock exam and a final project presented to the group; certificate of completion for participants who complete the labs and the project
OUTCOME	Participants can install, secure, upgrade, back up and troubleshoot a Kubernetes cluster from the command line and sit the CKA exam

PHASE 01

6 H

Cluster Installation & Lifecycle

Participants build a cluster the way the exam and production expect: a container runtime, kubeadm, a highly available control plane behind a load balancer. They then upgrade it one minor version and back up and restore etcd. Validated lab: a three-control-plane cluster upgraded without downtime, restored from an etcd snapshot.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Cluster Installation	kubeadm, highly available control plane, container runtime	Lecture + Lab	3 h
1.2	Cluster Lifecycle	Version upgrades, etcd backup and restore	Lecture + Lab	3 h

Security & Scheduling

Participants grant each user and workload only the access it needs, run containers with restrictive security contexts, then decide where pods run with taints and tolerations, affinity, resource requests and priority classes. Validated lab: a team namespace with least-privilege RBAC, and a workload pinned to dedicated nodes that keeps priority under pressure.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Security	RBAC, ServiceAccounts, security contexts	Lecture + Lab	3 h
2.2	Scheduling	Taints and tolerations, affinity, resource requests and limits, priority classes; recap of rollouts, ConfigMaps and Secrets, workload autoscaling	Lecture + Lab	3 h

PHASE 03

6 H

Networking, Storage & Extensions

Participants configure cluster networking from the CNI plugin to Services, CoreDNS, NetworkPolicies and the Gateway API, migrate an Ingress off the retired ingress-nginx, provision storage dynamically, and extend the cluster with Helm, Kustomize, custom resources and operators. Validated lab: an application reachable through the Gateway API, isolated by NetworkPolicies, with a dynamically provisioned volume.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Networking	CNI, Services, CoreDNS, NetworkPolicies, Gateway API, Ingress and migrating off ingress-nginx	Lecture + Lab	3 h
3.2	Storage	PersistentVolumes and PersistentVolumeClaims, access modes, reclaim policies, StorageClasses, dynamic provisioning	Lecture + Lab	1.5 h
3.3	Extending Kubernetes	Helm, Kustomize, extension interfaces (CNI, CSI, CRI), custom resource definitions and operators	Lecture + Lab	1.5 h

PHASE 04

6 H

Troubleshooting & Exam Preparation

Participants diagnose broken nodes, failing pods and network faults from logs, events and metrics, then sit a timed mock exam and learn the kubectl techniques that save time in a performance-based exam. Validated lab: a cluster with injected faults returned to a healthy state.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Troubleshooting	Nodes, pods, networking, logs and monitoring	Lecture + Lab	3 h
4.2	Exam Preparation	Timed mock exam, kubectl speed techniques	Lab	3 h

PHASE 05

6 H

Final Project

Each team builds a production-style cluster from a written brief on its lab server, reusing the phase labs, then repairs the faults injected by the trainer and presents its runbook.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Brief & Build	Read the brief, build a highly available cluster with kubeadm, secure it with RBAC and NetworkPolicies	Project	3 h
5.2	Operate & Document	Back up and restore etcd, upgrade the cluster, write the runbook	Project	1.5 h
5.3	Fault Repair & Review	Fix the faults injected by the trainer; questions and feedback	Project	1.5 h

FINAL PROJECT

A Production-Style Kubernetes Cluster

A highly available cluster built with kubeadm, secured with RBAC and NetworkPolicies, backed up and restored from etcd, upgraded one minor version, and returned to health after faults injected by the trainer.

STACK Kubernetes, kubeadm, containerd, etcd, a CNI plugin, CoreDNS, Envoy Gateway, MetalLB, local-path-provisioner, Helm, Kustomize, kubectl

DELIVERABLE Working cluster, runbook (install, backup, upgrade, recovery), fault report